

TRAINING DOCUMENT · SKILLCLOUD ACADEMY

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STRUCTURAL / GEOTECHNICAL FOUNDATION EVALUATION

Pacific Foundation Engineering · Robert Cheng, PE (Civil #C-68214) · Date: October 17, 2025

ENGINEERING INSPECTION REPORT

Property:	4827 Rolando Blvd, San Diego, CA 92115
Inspection Date:	October 17, 2025
Engineer:	Robert Cheng, PE, GE · Pacific Foundation Engineering
Scope of Work:	Specialized structural assessment of 18-inch horizontal crack on south foundation stem wall noted on home inspection.

ENGINEERING FINDINGS & DIAGNOSIS

1. Geotechnical & Soil Context:

The Rolando neighborhood is characterized by alluvial soil deposits subject to minor seasonal shrink-swell behavior. Laser level manometer readings across the floor slab indicate differential elevations within 0.35 inches across 40 feet, which falls well within acceptable ASTM residential tolerances.

2. Crack Characterization:

The 18-inch horizontal crack along the south stem wall represents **cosmetic settling and concrete curing shrinkage** dating back decades. **THERE IS NO STRUCTURAL DEFICIENCY, ROTATION, OR SIGNIFICANT SHEAR DISPLACEMENT.** The foundation continues to safely support the gravity and lateral loads of the superstructure.

3. Recommended Remediation:

To prevent moisture penetration and preserve reinforcement steel integrity, we recommend pressure epoxy injection sealing with structural urethane overcoat along the 18-inch fissure.

Remediation Cost Estimate: \$2,200.00 (Includes 5-year warranty on seal integrity)